

Alex Morgan

Data Engineer | Python, SQL, BigQuery & dbt | Toulouse

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Profile

Data engineer with two years of experience delivering reliable cloud pipelines, BigQuery transformations and operational analytics. Builds maintainable Python, SQL and dbt workflows with automated quality controls and monitoring, translating operational requirements into data products for technical and business teams.

Experience

Northwind Energy - Data Engineer 2024 - 2026

- Built and operated Python, SQL and Airflow pipelines integrating operational data into BigQuery.
- Developed dbt transformation models with automated data-quality controls and operational monitoring for daily datasets.
- Worked with operations and finance teams to define KPIs and deliver Looker reporting products.

UrbanFlow Analytics - Data Analyst Intern 2023 - 2024

- Prepared reusable SQL datasets and pandas analyses for mobility and service-quality reporting.
- Translated stakeholder questions into documented metrics, dashboards and validation checks.

Projects

Data quality observability | *dbt, Great Expectations, BigQuery, GitHub Actions*

- Automated freshness, schema and business-rule checks across BigQuery datasets with dbt, Great Expectations and CI reporting.
- Defined dataset ownership, alerting and traceable failure diagnostics so teams could investigate data incidents from source to dashboard.

Real-time mobility data platform | *Python, Kafka, Spark, Docker*

- Designed a Python-based streaming platform that validates, aggregates and exposes transport events for operational analytics.
- Added replayable integration tests, event-quality metrics and containerised local environments to support reliable delivery and future pipeline changes.

Energy demand forecasting | *Python, pandas, scikit-learn, MLflow*

- Built a reusable Python workflow for electricity-demand feature engineering, model training, MLflow experiment tracking and error analysis.
- Compared seasonal baselines with tree-based models and documented failure modes relevant to operational planning and workflow maintenance.

Skills

Data Engineering: Python, SQL, Airflow, Spark, Kafka, ETL/ELT

Cloud Platforms: GCP, BigQuery, Cloud Storage, Docker, Cloud Composer (transferable from Airflow)

Transformation & Quality: dbt, Great Expectations, data modelling, data quality & observability

Delivery & Operations: Git, GitHub Actions, CI/CD, MLflow, Looker

Education

Southwest Institute of Technology - Engineering degree, Data major 2021 - 2026

- Statistics, machine learning, databases, distributed systems and cloud architecture.

Technical University of Valencia - International semester 2024

- Data platforms, software engineering and project management.

Languages

French native | English C1

Interests

Energy systems | Open source